

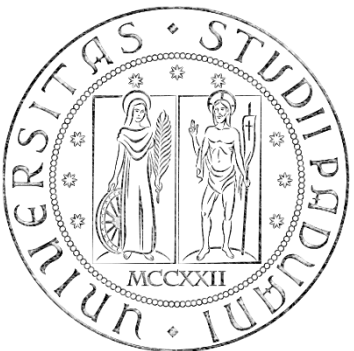
Biomedical Wearable Technologies  
for Healthcare and Wellbeing

# Exam and Project Discussion Q&A

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A.Y. 2025-2026

Giacomo Cappon



# Outline

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- **The theory exam**
- Project delivery instructions
- Project presentation guidelines

# Exam structure

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Written test of 90-minute duration:

- 4 open questions

Notes:

- The exam is in English, answers to be provided in English (English errors not penalized). Dictionary is not allowed.
- At the exam, the students cannot use any course material (slides, books, articles, etc.), nor any electronic device (e.g., smartphone, notebook, etc.). Only the pen and a calculator is allowed.
- The questions will focus on the course content presented during both the “theory” and the “laboratory” lessons except the material marked as **BONUS**
- In any case, no questions will require to write Dart/Flutter code.

# Mark of the theory exam

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- Open questions:
  - 0 to 5 points each
- Total score: up to 20 points

# Final mark

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- The written exam is passed if the student gets at least 12/20 points (which corresponds to 18/30).
- To get the final mark, students must pass both the written exam and the project discussion.
- The mark of the project is up to 12 (based on Q&A project discussion, different students in the same group can get different marks for the project).

*final mark = mark of the theory exam + mark of the project + symposium bonus*

# Examples of open questions

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Q1. Explain the RSA algorithm

Q2. Explain the role of the embedded processing unit

Q3. Provide a description of the following HTTP methods: GET, HEAD, POST, PUT, DELETE.

# Examples of open questions

Q4. Given the results of 5 SUS questionnaires collected by 5 participants evaluating an app A:

$SUS_A = 65, 80, 93, 78, 54$

- Calculate the confidence interval at 90% :  **$74 \pm 14.25$**
- Evaluate if an app B (evaluated by the same 5 participants) with SUS:

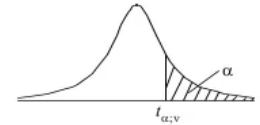
$SUS_B = 90, 90, 87, 95, 91$

can be said to be significantly more usable with a level of confidence  $\geq 0.05$ :

**$t = 2.30 \rightarrow \text{NO (if it was } \geq 0.1, \text{ YES)}$**

**Table of the Student's  $t$ -distribution**

The table gives the values of  $t_{\alpha;v}$  where  
 $\Pr(T_v > t_{\alpha;v}) = \alpha$ , with  $v$  degrees of freedom



| $\alpha \backslash v$ | 0.1   | 0.05  | 0.025  | 0.01   | 0.005  | 0.001   | 0.0005  |
|-----------------------|-------|-------|--------|--------|--------|---------|---------|
| 1                     | 3.078 | 6.314 | 12.076 | 31.821 | 63.657 | 318.310 | 636.620 |
| 2                     | 1.886 | 2.920 | 4.303  | 6.965  | 9.925  | 22.326  | 31.598  |
| 3                     | 1.638 | 2.353 | 3.182  | 4.541  | 5.841  | 10.213  | 12.924  |
| 4                     | 1.533 | 2.132 | 2.776  | 3.747  | 4.604  | 7.173   | 8.610   |
| 5                     | 1.476 | 2.015 | 2.571  | 3.365  | 4.032  | 5.893   | 6.869   |
| 6                     | 1.440 | 1.943 | 2.447  | 3.143  | 3.707  | 5.208   | 5.959   |
| 7                     | 1.415 | 1.895 | 2.365  | 2.998  | 3.499  | 4.785   | 5.408   |
| 8                     | 1.397 | 1.860 | 2.306  | 2.896  | 3.355  | 4.501   | 5.041   |
| 9                     | 1.383 | 1.833 | 2.262  | 2.821  | 3.250  | 4.297   | 4.781   |
| 10                    | 1.372 | 1.812 | 2.228  | 2.764  | 3.169  | 4.144   | 4.587   |
| 11                    | 1.363 | 1.796 | 2.201  | 2.718  | 3.106  | 4.025   | 4.437   |
| 12                    | 1.356 | 1.782 | 2.179  | 2.681  | 3.055  | 3.930   | 4.318   |
| 13                    | 1.350 | 1.771 | 2.160  | 2.650  | 3.012  | 3.852   | 4.221   |
| 14                    | 1.345 | 1.761 | 2.145  | 2.624  | 2.977  | 3.787   | 4.140   |
| 15                    | 1.341 | 1.753 | 2.131  | 2.602  | 2.947  | 3.733   | 4.073   |
| 16                    | 1.337 | 1.746 | 2.120  | 2.583  | 2.921  | 3.686   | 4.015   |
| 17                    | 1.333 | 1.740 | 2.110  | 2.567  | 2.898  | 3.646   | 3.965   |
| 18                    | 1.330 | 1.734 | 2.101  | 2.552  | 2.878  | 3.610   | 3.922   |
| 19                    | 1.328 | 1.729 | 2.093  | 2.539  | 2.861  | 3.579   | 3.883   |
| 20                    | 1.325 | 1.725 | 2.086  | 2.528  | 2.845  | 3.552   | 3.850   |
| 21                    | 1.323 | 1.721 | 2.080  | 2.518  | 2.831  | 3.527   | 3.819   |
| 22                    | 1.321 | 1.717 | 2.074  | 2.508  | 2.819  | 3.505   | 3.792   |
| 23                    | 1.319 | 1.714 | 2.069  | 2.500  | 2.807  | 3.485   | 3.767   |
| 24                    | 1.318 | 1.711 | 2.064  | 2.492  | 2.797  | 3.467   | 3.745   |
| 25                    | 1.316 | 1.708 | 2.060  | 2.485  | 2.787  | 3.450   | 3.725   |
| 26                    | 1.315 | 1.706 | 2.056  | 2.479  | 2.779  | 3.435   | 3.707   |
| 27                    | 1.314 | 1.703 | 2.052  | 2.473  | 2.771  | 3.421   | 3.690   |
| 28                    | 1.313 | 1.701 | 2.048  | 2.467  | 2.763  | 3.408   | 3.674   |
| 29                    | 1.311 | 1.699 | 2.045  | 2.462  | 2.756  | 3.396   | 3.659   |
| 30                    | 1.310 | 1.697 | 2.042  | 2.457  | 2.750  | 3.385   | 3.646   |
| 40                    | 1.303 | 1.684 | 2.021  | 2.423  | 2.704  | 3.307   | 3.551   |
| 60                    | 1.296 | 1.671 | 2.000  | 2.390  | 2.660  | 3.232   | 3.460   |
| 120                   | 1.289 | 1.658 | 1.980  | 2.358  | 2.617  | 3.160   | 3.373   |
| $\infty$              | 1.282 | 1.645 | 1.960  | 2.326  | 2.576  | 3.090   | 3.291   |

# Outline

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- The theory exam
- **Project delivery instructions**
- Project presentation guidelines



# General info

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- In the following, we reported the instructions on how to deliver your project. In a nutshell, you must prepare and submit a **.pdf file** containing
  - the number of your group and the name of each group participant
  - the link to a public GitHub repository containing a "valid" version of the code
- Since the project must be delivered through GitHub, code delivered via mail will be ignored
- "Valid" code must be in the **master** (or **main**) **branch** and must have been committed no later than the **12/07/2026**, and no later than the **day before the project discussion date**.

# Example scenario

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- "I belong to Group 50 composed by me, Martina Vettoretti, and Luca Cossu"
- "My code is located in my public repository named bwthw"

# Step 1: Fill the group info

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- Create a .docx file
- Fill the group info at the very top of the file, e.g.:

## Group 50

### Group members:

| ID      | Name               | Mail                                                                         |
|---------|--------------------|------------------------------------------------------------------------------|
| 1234567 | Giacomo Cappon     | <a href="mailto:giacomo.cappon@unipd.it">giacomo.cappon@unipd.it</a>         |
| 1234568 | Martina Vettoretti | <a href="mailto:martina.vettoretti@unipd.it">martina.vettoretti@unipd.it</a> |
| 1234569 | Luca Cossu         | <a href="mailto:luca.cossu@phd.unipd.it">luca.cossu@phd.unipd.it</a>         |

# Step 2: Make your repository public

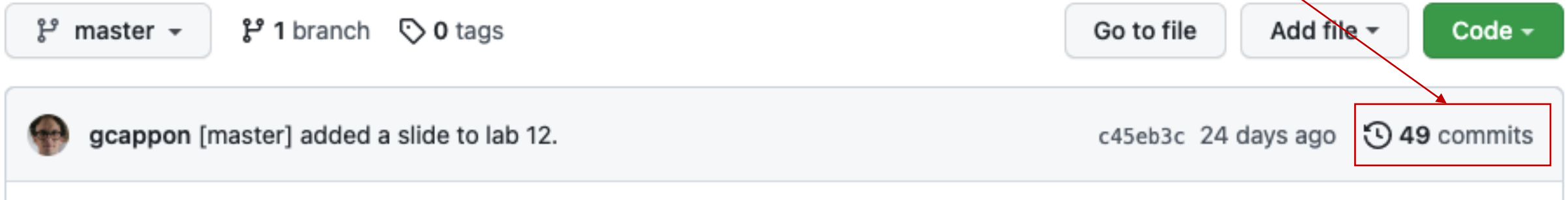
- First, make sure your repository is public. This is true if, in GitHub, a “public” label is present near your repo name:
- If, instead, the label says “private”, follow these steps to make it public:

- Go to “Settings”
- Scroll down till the “Danger Zone”
- Click on “Change visibility”
- Select “Make public”
- Type the name of the repo to confirm
- Click on the button

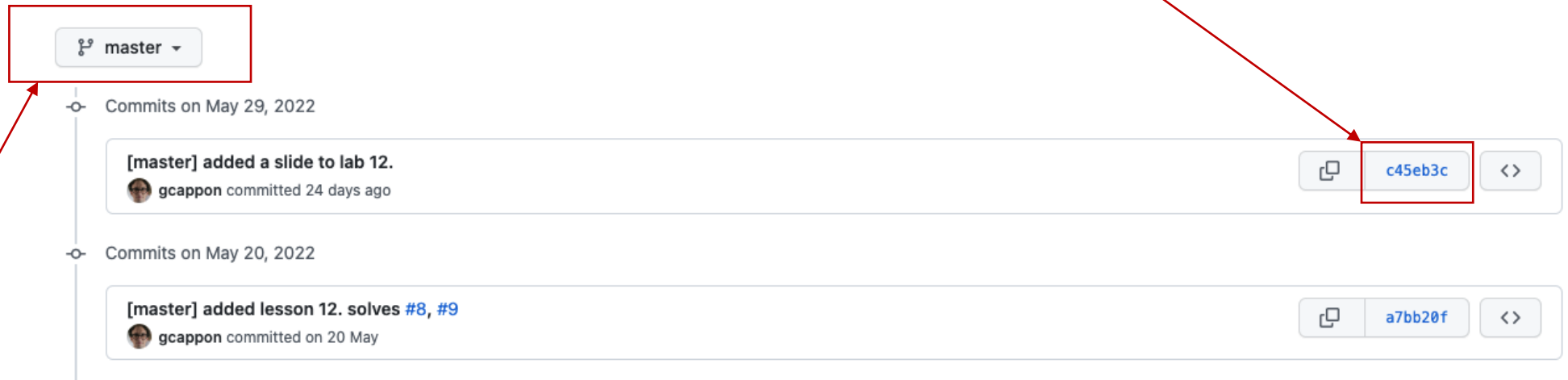
The screenshot shows the GitHub repository settings page for 'gcappon / bwthw'. The repository is currently marked as 'Public'. The 'Settings' tab is selected in the top navigation bar. In the 'Danger Zone', the 'Change repository visibility' section is highlighted. A modal dialog titled 'Change repository visibility' is open, showing a warning: 'Warning: this is a potentially destructive action.' Below the warning, the 'Make public' option is selected with a radio button, and the text 'This repository is currently public.' is displayed. The 'Make private' option is also visible. At the bottom of the modal, there is a confirmation prompt: 'Please type gcappon/bwthw to confirm.' A text input field contains 'gcappon/bwthw', and a button labeled 'I understand, change repository visibility.' is at the bottom.

# Step 3: Locate the link to a valid code version

- From the home page of your GitHub repo, click on the “history” link



- Click on the link of the version of the code that you want to submit



- Note: be sure that your are in the master (or main) branch!

# Step 4: Put the link in the .docx file

- Copy and paste the link in the url bar of your browser in the .docx file below the group info, e.g.:

← → ↻ 🏠 [github.com/gcappon/bwthw/commit/c45eb3c20d26fe17af9c05c4fb94e883d0404929](https://github.com/gcappon/bwthw/commit/c45eb3c20d26fe17af9c05c4fb94e883d0404929)



## Group 50

Group members:

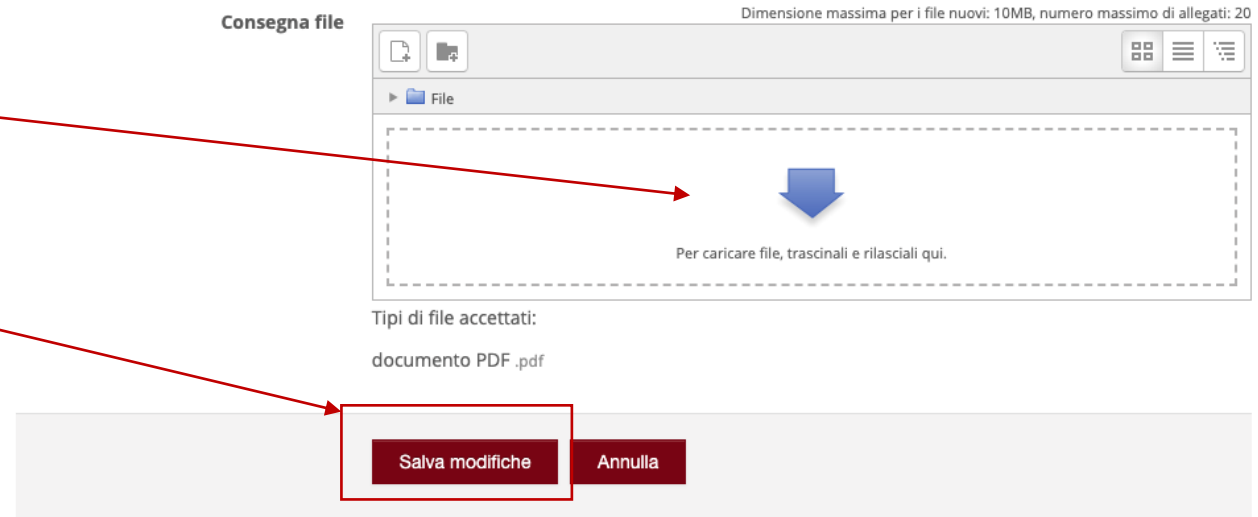
| ID      | Name               | Mail                                                                         |
|---------|--------------------|------------------------------------------------------------------------------|
| 1234567 | Giacomo Cappon     | <a href="mailto:giacomo.cappon@unipd.it">giacomo.cappon@unipd.it</a>         |
| 1234568 | Martina Vettoretti | <a href="mailto:martina.vettoretti@unipd.it">martina.vettoretti@unipd.it</a> |
| 1234569 | Luca Cossu         | <a href="mailto:luca.cossu@phd.unipd.it">luca.cossu@phd.unipd.it</a>         |

Link to the GitHub repository:

<https://github.com/gcappon/bwthw/commit/c45eb3c20d26fe17af9c05c4fb94e883d0404929>

# Step 5: Submit the .pdf to moodle

- Export your .docx file to .pdf
- Then, go to moodle. In the “Project discussion – Project delivery” section locate “Project delivery”:
- Click on the delivery button:
- Upload the .pdf file
- Save and that's it!



# Outline

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- The theory exam
- Project delivery instructions
- **Project presentation guidelines**



# Rationale

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- In the following we reported a brief guide that contains some **hints** on what to cover during your project discussion. As such, consider this document as a guide rather than something to “fill” with words and images.
- You are free to use the template your prefer
- The skeleton proposed here is just a simple guideline. Feel free to change it according to your needs.

# Discussion structure – General info

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- You have **25 minutes max** to present your project followed by ~20 min of Q&A. Discussions of more than 25 minutes will be penalized.
- The presentation must include a PowerPoint presentation followed by a live demonstration of your app working. Both the presentation and the demo must fit the 25 minutes.
- The discussion must be in English.
- The Q&A session will be in English.

# Discussion structure – (Possible) skeleton

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- A possible skeleton for your discussion can be:
  - **Background/Target**
  - **Problem to be solved/Feature to be provided to the public –**
  - **Your solution**
  - **Core app functionalities**
  - **Original part**
  - **Project management flavours**
  - **Special implementation decisions**
  - **Closing remarks and future developments**
  - **Live demo** - *Show (via emulator or physical device) your app working.*

# Discussion suggestions

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- Try to motivate every choice.
- Consider in advance how much time you will allocate to the PowerPoint presentation and how much time for the live demo.
- Prepare the live demo. You have limited time so you must know what to "tap", to show, to do ,....
- Be sure that your app is working when you run the demo.